Smart Waste Collection

An individual, data-driven Monte Carlo simulation of collection-threshold decisions

Executive summary

Smart-bin sensors record discrete fullness states over time. This project transforms historical observations into state-dependent probabilistic inputs and uses Monte Carlo simulation to compare collection-trigger thresholds of 6, 8 and 10. The core decision is a trade-off: collecting earlier increases collection frequency but reduces exposure to highly full bins.

Across 1,000 simulated 365-day years per base scenario, threshold 6 averages 75.30 collections and 9.18% high-fullness days; threshold 8 averages 62.84 collections and 17.22%; threshold 10 averages 24.93 collections and 66.42%. The findings are simulation outcomes, not observed operating performance or an optimization of real-world cost.

1. Problem, motivation and scope

Fixed or poorly calibrated collection triggers may lead either to unnecessary collection activity or prolonged high-fullness conditions. A smart city can use sensor histories to examine these competing consequences before choosing a rule.

Research question: How do alternative collection thresholds change expected collection frequency and high-fullness exposure when daily accumulation is uncertain?

In scope: data understanding, quality audit, daily state transitions, empirical random-variable construction, simulation, scenario comparison and sensitivity testing.

Out of scope: vehicle routing, collection schedules, truck capacity, costs, emissions and causal estimation of historical policy performance.

2. Data and interpretation

Source: Wyndham smart-bin fill-level historical GeoJSON (wyndham_smartbin_filllevel.json), Fields include bin identifier, date, fullness state, status/reason, configured threshold and location. The original source contains 31,427 rows and 33 unique bins.

Fullness values are discrete states 0, 2, 4, 6, 8 and 10; they are not assumed to be percentages. State 10 is the maximum observed state, not a verified overflow event. ALERT has no confirmed operational interpretation. A fall to zero is a potential emptying/reset proxy, not verified collection evidence.

3. Data audit and preparation

Audit identified 766 exact duplicate records (2.44%), removed to obtain 30,661 master records. Four records from two ambiguous same-bin/date combinations were retained in the master but excluded from transition modelling. Across the 1,043-day calendar span, 966 dates were observed and 77 were absent; missing days were not filled with invented readings.

Bin 1511202 has 181 records, all at fullness state 0 with ALERT status. It was retained for traceability but excluded from behaviour modelling. This left 30,476 observations across 32 modelling bins. Only pairs exactly one day apart were used, producing 29,999 daily transitions and 4,514 potential emptying events.

Reproducible data products include smart_bin_processed.csv, smart_bin_daily_transitions.csv, smart_bin_emptying_events.csv and smart_bin_transition_probabilities.csv.

4. Random variables and simulation design

RV1 represents whether a bin accumulates waste on the next day, with probabilities conditioned on its current fullness state. RV2 represents the size of an increase conditional on accumulation. Empirical probabilities are derived from observed transitions rather than imposing an arbitrary normal distribution.

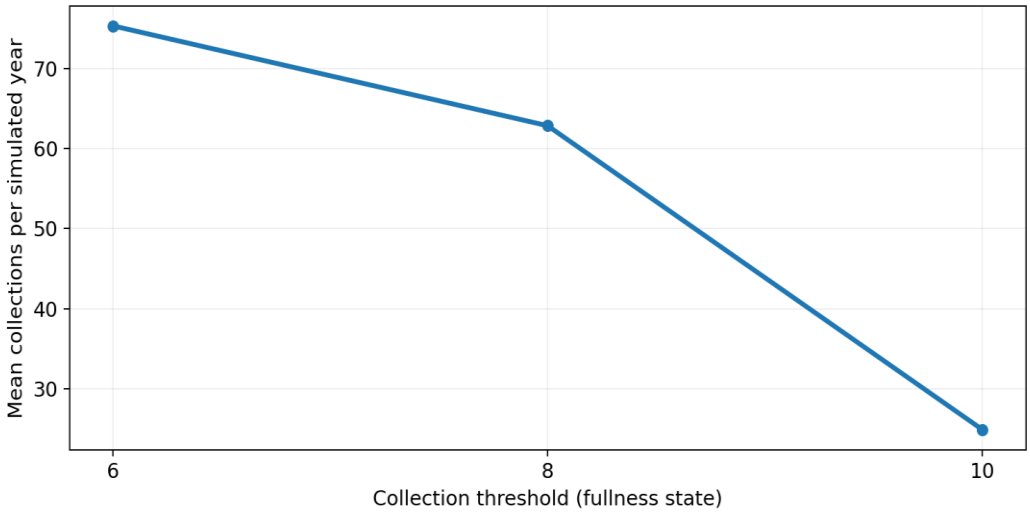
Each simulated day: begin at the current fullness state; sample accumulation occurrence; if it occurs, sample increase magnitude; update fullness subject to the model state rules; apply the selected threshold; when triggered, count a collection and reset the state. The model is a representative pooled-bin simulation, not a separate calibrated forecast for every physical bin.

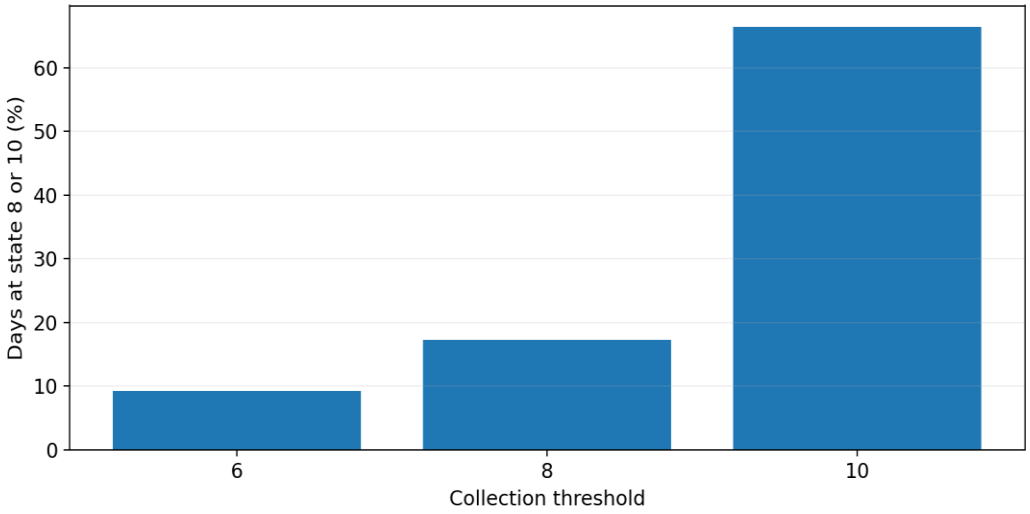
Baseline: initial state 0, threshold 8, 365 days per run, 1,000 replications, base seed 42. The supplied baseline summary reports 62.836 mean annual collections and 4.997 mean days at state 10.

5. Scenario experiment and results

Three collection thresholds (6, 8, 10) were evaluated under the same base accumulation model. The outputs track mean annual collections, maximum-state days, average fullness, fullness at collection and percentage of days at state 8 or 10.

Threshold	Collections/year	Days at state 10	High-fullness days %	Mean fullness
6	75.30	3.59	9.18%	2.37
8	62.84	5.00	17.22%	3.16
10	24.93	24.93	66.42%	6.16

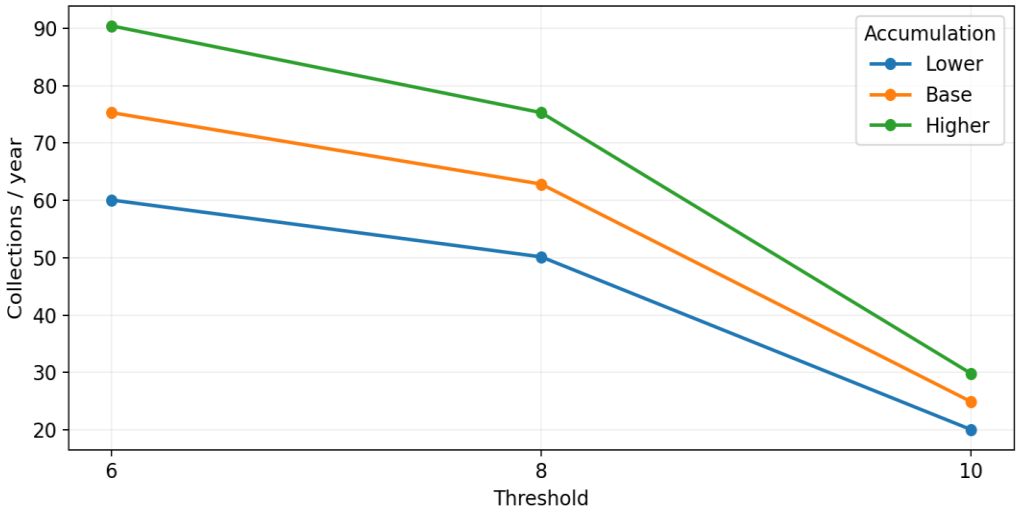




6. Sensitivity analysis

The probability of accumulation was stress-tested at −20% (Lower), unchanged (Base) and +20% (Higher), with the no-change probability adjusted to preserve total probability. Increase magnitude was held fixed. The ±20% shifts are assumptions for robustness testing, not data-derived forecasts.

The 3 × 3 design contains nine experiments, each with 1,000 runs of 365 days: 3,285,000 simulated bin-days in total. Collection counts rise with accumulation for all thresholds, but threshold ordering remains 6 > 8 > 10 for collections and 6 < 8 < 10 for high-fullness exposure.



7. Interpretation and decision implications

Threshold 6 prioritizes reducing high-fullness exposure but requires more collections. Threshold 10 prioritizes fewer collections but substantially increases high-fullness exposure. Threshold 8 occupies the middle ground in this experiment. No threshold is universally optimal without explicit service-level priorities and cost weights.

The analysis demonstrates a decision-support workflow: observed sensor behaviour → audited daily transitions → empirical uncertainty → simulated alternatives → robustness testing → conditional policy guidance.

8. Validation, limitations and next steps

The supplied simulation outputs and notebooks support checking logic, baseline reproducibility and scenario comparisons. Results should not be treated as operationally validated against actual collection logs, because such logs are not included.

Limitations: unknown semantics of ALERT and fullness scale; inferred rather than observed emptying; pooled state-dependent behaviour; simplified instantaneous threshold-triggered reset; no routing, time-to-service, costs or emissions; limited sensitivity range.

Next steps: confirm sensor metadata and collection event logs; calibrate and validate per-bin behaviour; model service delays and capacity; add cost/service-level objectives; then test policy performance in a controlled field pilot.

All outcomes in this document refer to the supplied experiment results. Descriptions of earlier team/member responsibilities in the original narrative have deliberately not been carried into this individual-project document.